

LLaMA.cpp HTTP Server

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/*
notes: 杨小兵-2024-12-16

1、这个项目是llama.cpp项目的子项目HTTP Server
2、这个子项目作为一个llama.cpp的前端从而使得更容易使用后端的模型
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```

Fast, lightweight, pure C/C++ HTTP server based on [httplib](#), [nlohmann::json](#) and **llama.cpp**.

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1、快速的、轻量级的、纯C/C++ HTTP server
2、这个HTTP server依赖httplib、json、llama.cpp这三个库
3、上述提到的httplib是我没有接触过的
4、llama.cpp是这个HTTP server的计算后端
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```

Set of LLM REST APIs and a simple web front end to interact with llama.cpp.

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1、组成部分
  1.1 一组 LLM REST APIs
  1.2 一个用于与 llama.cpp 交互的简单 Web 前端
2、对HTTP server的简单介绍
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```

Features:

- LLM inference of F16 and quantized models on GPU and CPU
- [OpenAI API](#) compatible chat completions and embeddings routes
- Reranking endpoint (WIP: <https://github.com/ggerranov/llama.cpp/pull/9510>)
- Parallel decoding with multi-user support
- Continuous batching
- Multimodal (wip)
- Monitoring endpoints
- Schema-constrained JSON response format

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1、支持在GPU和CPU上进行大规模语言模型（LLM）的推理，使用F16（16位浮点）和量化模型。F16模型通过减少每个参数的位数来加快计算速度并降低内存占用，而量化模型则通过将模型参数压缩为更低的精度（例如INT8）来进一步提高效率
2、提供与OpenAI API兼容的聊天补全（chat completions）和嵌入（embeddings）接口。这意味着开发者可以使用与OpenAI相同的API端点和请求格式来集成和调用该服务，简化了迁移和集成过程
3、提供一个重新排序（reranking）的接口端点，用于对生成的结果进行二次排序，以提高结果的相关性和准确性。目前该功能仍在开发中（Work In Progress, WIP），具体实现可参考提供的GitHub拉取请求链接
4、支持并行解码，允许多个用户同时进行推理请求。这意味着系统可以同时处理多个查询，提高响应速度和吞吐量，适用于高并发的应用场景
5、实现持续的批处理机制，将多个推理请求合并为一个批次进行处理，以提高计算资源的利用率和推理效率。这在处理大量短请求时尤其有效，能够减少单个请求的延迟
6、支持多模态功能，即处理和理解多种类型的数据输入，如文本、图像、音频等。目前该功能仍在开发中，未来将能够处理和生成多种形式的内容
7、提供监控端点，用于实时跟踪和监控系统的运行状态、性能指标和资源使用情况。开发者可以通过这些端点获取系统健康状况，进行故障排查和性能优化。
8、采用基于模式（Schema）约束的JSON响应格式，确保API返回的数据结构符合预定义的规范。这有助于提高数据的一致性和可靠性，便于客户端解析和处理。

Chat Completions：聊天补全功能，指的是模型根据对话上下文生成相应的回复，用于构建对话系统或聊天机器人。
OpenAI API：OpenAI提供的应用程序编程接口，允许开发者通过HTTP请求与其语言模型交互，执行文本生成、嵌入生成等功能。兼容性意味着该服务可以无缝替代或集成OpenAI的API。
Embeddings：嵌入是将文本或其他数据转换为高维向量表示的技术，这些向量可以捕捉语义信息，广泛应用于搜索、推荐系统、相似性计算等领域。
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```

The project is under active development, and we are [looking for feedback and contributors](#).

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1、这个项目还处于积极开发状态，想要获取一些反馈和贡献者
2、后面将会进行积极的开发和维护
*/
```

1 Usage

```
/*
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1、贡献者需要注意：下列列表是通过llama-gen-docs生成得到的
2、通过名为llama-gen-docs可执行文件得到的
*/
```

Common params

Argument	Explanation
<code>-h, --help, --usage</code>	print usage and exit
<code>--version</code>	show version and build info
<code>--verbose-prompt</code>	print a verbose prompt before generation (default: false)
<code>-t, --threads N</code>	number of threads to use during generation (default: -1) (env: LLAMA_ARG_THREADS)
<code>-tb, --threads-batch N</code>	number of threads to use during batch and prompt processing (default: same as --threads)
<code>-C, --cpu-mask M</code>	CPU affinity mask: arbitrarily long hex. Complements cpu-range (default: "")
<code>-Cr, --cpu-range lo-hi</code>	range of CPUs for affinity. Complements --cpu-mask
<code>--cpu-strict <0\ 1></code>	use strict CPU placement (default: 0)
<code>--prio N</code>	set process/thread priority : 0-normal, 1-medium, 2-high, 3-realtime (default: 0)
<code>--poll <0...100></code>	use polling level to wait for work (0 - no polling, default: 50)
<code>-Cb, --cpu-mask-batch M</code>	CPU affinity mask: arbitrarily long hex. Complements cpu-range-batch (default: same as --cpu-mask)
<code>-Crb, --cpu-range-batch lo-hi</code>	ranges of CPUs for affinity. Complements --cpu-mask-batch
<code>--cpu-strict-batch <0\ 1></code>	use strict CPU placement (default: same as --cpu-strict)
<code>--prio-batch N</code>	set process/thread priority : 0-normal, 1-medium, 2-high, 3-realtime (default: 0)
<code>--poll-batch <0\ 1></code>	use polling to wait for work (default: same as --poll)
<code>-c, --ctx-size N</code>	size of the prompt context (default: 4096, 0 = loaded from model) (env: LLAMA_ARG_CTX_SIZE)
<code>-n, --predict, --n-predict N</code>	number of tokens to predict (default: -1, -1 = infinity, -2 = until context filled) (env: LLAMA_ARG_N_PREDICT)
<code>-b, --batch-size N</code>	logical maximum batch size (default: 2048) (env: LLAMA_ARG_BATCH)
<code>-ub, --ubatch-size N</code>	physical maximum batch size (default: 512) (env: LLAMA_ARG_UBATCH)
<code>--keep N</code>	number of tokens to keep from the initial prompt (default: 0, -1 = all)

Argument	Explanation
<code>-fa, --flash-attn</code>	enable Flash Attention (default: disabled) (env: LLAMA_ARG_FLASH_ATTN)
<code>-p, --prompt PROMPT</code>	prompt to start generation with
<code>--no-perf</code>	disable internal libllama performance timings (default: false) (env: LLAMA_ARG_NO_PERF)
<code>-f, --file FNAME</code>	a file containing the prompt (default: none)
<code>-bf, --binary-file FNAME</code>	binary file containing the prompt (default: none)
<code>-e, --escape</code>	process escapes sequences (\n, \r, \t, ', ", \) (default: true)
<code>--no-escape</code>	do not process escape sequences
<code>--rope-scaling {none,linear,yarn}</code>	RoPE frequency scaling method, defaults to linear unless specified by the model (env: LLAMA_ARG_ROPE_SCALING_TYPE)
<code>--rope-scale N</code>	RoPE context scaling factor, expands context by a factor of N (env: LLAMA_ARG_ROPE_SCALE)
<code>--rope-freq-base N</code>	RoPE base frequency, used by NTK-aware scaling (default: loaded from model) (env: LLAMA_ARG_ROPE_FREQ_BASE)
<code>--rope-freq-scale N</code>	RoPE frequency scaling factor, expands context by a factor of 1/N (env: LLAMA_ARG_ROPE_FREQ_SCALE)
<code>--yarn-orig-ctx N</code>	YaRN: original context size of model (default: 0 = model training context size) (env: LLAMA_ARG_YARN_ORIG_CTX)
<code>--yarn-ext-factor N</code>	YaRN: extrapolation mix factor (default: -1.0, 0.0 = full interpolation) (env: LLAMA_ARG_YARN_EXT_FACTOR)
<code>--yarn-attn-factor N</code>	YaRN: scale sqrt(t) or attention magnitude (default: 1.0) (env: LLAMA_ARG_YARN_ATTN_FACTOR)
<code>--yarn-beta-slow N</code>	YaRN: high correction dim or alpha (default: 1.0) (env: LLAMA_ARG_YARN_BETA_SLOW)
<code>--yarn-beta-fast N</code>	YaRN: low correction dim or beta (default: 32.0) (env: LLAMA_ARG_YARN_BETA_FAST)
<code>-dkvc, --dump-kv-cache</code>	verbose print of the KV cache
<code>-nkvo, --no-kv-offload</code>	disable KV offload (env: LLAMA_ARG_NO_KV_OFFLOAD)

Argument	Explanation
<code>-ctk, --cache-type-k TYPE</code>	KV cache data type for K allowed values: f32, f16, bf16, q8_0, q4_0, q4_1, iq4_nl, q5_0, q5_1 (default: f16) (env: LLAMA_ARG_CACHE_TYPE_K)
<code>-ctv, --cache-type-v TYPE</code>	KV cache data type for V allowed values: f32, f16, bf16, q8_0, q4_0, q4_1, iq4_nl, q5_0, q5_1 (default: f16) (env: LLAMA_ARG_CACHE_TYPE_V)
<code>-dt, --defrag-thold N</code>	KV cache defragmentation threshold (default: 0.1, < 0 - disabled) (env: LLAMA_ARG_DEFRAG_THOLD)
<code>-np, --parallel N</code>	number of parallel sequences to decode (default: 1) (env: LLAMA_ARG_N_PARALLEL)
<code>--mlock</code>	force system to keep model in RAM rather than swapping or compressing (env: LLAMA_ARG_MLOCK)
<code>--no-mmap</code>	do not memory-map model (slower load but may reduce pageouts if not using mlock) (env: LLAMA_ARG_NO_MMAP)
<code>--numa TYPE</code>	attempt optimizations that help on some NUMA systems - distribute: spread execution evenly over all nodes - isolate: only spawn threads on CPUs on the node that execution started on - numactl: use the CPU map provided by numactl if run without this previously, it is recommended to drop the system page cache before using this see https://github.com/ggerganov/llama.cpp/issues/1437 (env: LLAMA_ARG_NUMA)
<code>-dev, --device <dev1,dev2,...></code>	comma-separated list of devices to use for offloading (none = don't offload) use --list-devices to see a list of available devices (env: LLAMA_ARG_DEVICE)
<code>--list-devices</code>	print list of available devices and exit
<code>-ngl, --gpu-layers, --n-gpu-layers N</code>	number of layers to store in VRAM (env: LLAMA_ARG_N_GPU_LAYERS)
<code>-sm, --split-mode {none,layer,row}</code>	how to split the model across multiple GPUs, one of: - none: use one GPU only - layer (default): split layers and KV across GPUs - row: split rows across GPUs (env: LLAMA_ARG_SPLIT_MODE)
<code>-ts, --tensor-split N0,N1,N2,...</code>	fraction of the model to offload to each GPU, comma-separated list of proportions, e.g. 3,1 (env: LLAMA_ARG_TENSOR_SPLIT)

Argument	Explanation
<code>-mg, --main-gpu INDEX</code>	the GPU to use for the model (with split-mode = none), or for intermediate results and KV (with split-mode = row) (default: 0) (env: LLAMA_ARG_MAIN_GPU)
<code>--check-tensors</code>	check model tensor data for invalid values (default: false)
<code>--override-kv KEY=TYPE:VALUE</code>	advanced option to override model metadata by key. may be specified multiple times. types: int, float, bool, str. example: <code>--override-kv tokenizer.ggml.add_bos_token=bool:false</code>
<code>--lora FNAME</code>	path to LoRA adapter (can be repeated to use multiple adapters)
<code>--lora-scaled FNAME SCALE</code>	path to LoRA adapter with user defined scaling (can be repeated to use multiple adapters)
<code>--control-vector FNAME</code>	add a control vector note: this argument can be repeated to add multiple control vectors
<code>--control-vector-scaled FNAME SCALE</code>	add a control vector with user defined scaling SCALE note: this argument can be repeated to add multiple scaled control vectors
<code>--control-vector-layer-range START END</code>	layer range to apply the control vector(s) to, start and end inclusive
<code>-m, --model FNAME</code>	model path (default: <code>models/\$filename</code> with filename from <code>--hf-file</code> or <code>--model-url</code> if set, otherwise <code>models/7B/ggml-model-f16.gguf</code>) (env: LLAMA_ARG_MODEL)
<code>-mu, --model-url MODEL_URL</code>	model download url (default: unused) (env: LLAMA_ARG_MODEL_URL)
<code>-hfr, --hf-repo REPO</code>	Hugging Face model repository (default: unused) (env: LLAMA_ARG_HF_REPO)
<code>-hff, --hf-file FILE</code>	Hugging Face model file (default: unused) (env: LLAMA_ARG_HF_FILE)
<code>-hft, --hf-token TOKEN</code>	Hugging Face access token (default: value from HF_TOKEN environment variable) (env: HF_TOKEN)
<code>--log-disable</code>	Log disable
<code>--log-file FNAME</code>	Log to file
<code>--log-colors</code>	Enable colored logging (env: LLAMA_LOG_COLORS)
<code>-v, --verbose, --log-verbose</code>	Set verbosity level to infinity (i.e. log all messages, useful for debugging)

Argument	Explanation
<code>-lv, --verbosity, --log-verbosity N</code>	Set the verbosity threshold. Messages with a higher verbosity will be ignored. (env: LLAMA_LOG_VERBOSITY)
<code>--log-prefix</code>	Enable prefix in log messages (env: LLAMA_LOG_PREFIX)
<code>--log-timestamps</code>	Enable timestamps in log messages (env: LLAMA_LOG_TIMESTAMP)

Sampling params

Argument	Explanation
<code>--samplers</code> SAMPLERS	samplers that will be used for generation in the order, separated by ';' (default: dry;top_k;typ_p;top_p;min_p;xtc;temperature)
<code>-s, --seed SEED</code>	RNG seed (default: -1, use random seed for -1)
<code>--sampling-seq</code> SEQUENCE	simplified sequence for samplers that will be used (default: dkypmxt)
<code>--ignore-eos</code>	ignore end of stream token and continue generating (implies --logit-bias EOS-inf)
<code>--penalize-nl</code>	penalize newline tokens (default: false)
<code>--temp N</code>	temperature (default: 0.8)
<code>--top-k N</code>	top-k sampling (default: 40, 0 = disabled)
<code>--top-p N</code>	top-p sampling (default: 0.9, 1.0 = disabled)
<code>--min-p N</code>	min-p sampling (default: 0.1, 0.0 = disabled)
<code>--xtc-probability</code> N	xtc probability (default: 0.0, 0.0 = disabled)
<code>--xtc-threshold N</code>	xtc threshold (default: 0.1, 1.0 = disabled)
<code>--typical N</code>	locally typical sampling, parameter p (default: 1.0, 1.0 = disabled)
<code>--repeat-last-n N</code>	last n tokens to consider for penalize (default: 64, 0 = disabled, -1 = ctx_size)
<code>--repeat-penalty N</code>	penalize repeat sequence of tokens (default: 1.0, 1.0 = disabled)
<code>--presence-penalty</code> N	repeat alpha presence penalty (default: 0.0, 0.0 = disabled)
<code>--frequency-penalty</code> N	repeat alpha frequency penalty (default: 0.0, 0.0 = disabled)
<code>--dry-multiplier N</code>	set DRY sampling multiplier (default: 0.0, 0.0 = disabled)
<code>--dry-base N</code>	set DRY sampling base value (default: 1.75)

Argument	Explanation
<code>--dry-allowed-length N</code>	set allowed length for DRY sampling (default: 2)
<code>--dry-penalty-last-n N</code>	set DRY penalty for the last n tokens (default: -1, 0 = disable, -1 = context size)
<code>--dry-sequence-breaker STRING</code>	add sequence breaker for DRY sampling, clearing out default breakers ("\\n", ':', '"', '*') in the process; use "none" to not use any sequence breakers
<code>--dynatemp-range N</code>	dynamic temperature range (default: 0.0, 0.0 = disabled)
<code>--dynatemp-exp N</code>	dynamic temperature exponent (default: 1.0)
<code>--mirostat N</code>	use Mirostat sampling. Top K, Nucleus and Locally Typical samplers are ignored if used. (default: 0, 0 = disabled, 1 = Mirostat, 2 = Mirostat 2.0)
<code>--mirostat-lr N</code>	Mirostat learning rate, parameter eta (default: 0.1)
<code>--mirostat-ent N</code>	Mirostat target entropy, parameter tau (default: 5.0)
<code>-l, --logit-bias TOKEN_ID(+/-)BIAS</code>	modifies the likelihood of token appearing in the completion, i.e. <code>--logit-bias 15043+1</code> to increase likelihood of token ' Hello', or <code>--logit-bias 15043-1</code> to decrease likelihood of token ' Hello'
<code>--grammar GRAMMAR</code>	BNF-like grammar to constrain generations (see samples in grammars/ dir) (default: "")
<code>--grammar-file FNAME</code>	file to read grammar from
<code>-j, --json-schema SCHEMA</code>	JSON schema to constrain generations (https://json-schema.org/), e.g. <code>{}</code> for any JSON object For schemas w/ external \$refs, use <code>--grammar + example/json_schema_to_grammar.py</code> instead

Example-specific params

Argument	Explanation
<code>--no-context-shift</code>	disables context shift on infinite text generation (default: disabled) (env: LLAMA_ARG_NO_CONTEXT_SHIFT)
<code>-sp, --special</code>	special tokens output enabled (default: false)
<code>--no-warmup</code>	skip warming up the model with an empty run
<code>--spm-infill</code>	use Suffix/Prefix/Middle pattern for infill (instead of Prefix/Suffix/Middle) as some models prefer this. (default: disabled)
<code>--pooling {none,mean,cls,last,rank}</code>	pooling type for embeddings, use model default if unspecified (env: LLAMA_ARG_POOLING)

Argument	Explanation
<code>-cb, --cont-batching</code>	enable continuous batching (a.k.a dynamic batching) (default: enabled) (env: LLAMA_ARG_CONT_BATCHING)
<code>-nocb, --no-cont-batching</code>	disable continuous batching (env: LLAMA_ARG_NO_CONT_BATCHING)
<code>-a, --alias STRING</code>	set alias for model name (to be used by REST API) (env: LLAMA_ARG_ALIAS)
<code>--host HOST</code>	ip address to listen (default: 127.0.0.1) (env: LLAMA_ARG_HOST)
<code>--port PORT</code>	port to listen (default: 8080) (env: LLAMA_ARG_PORT)
<code>--path PATH</code>	path to serve static files from (default:) (env: LLAMA_ARG_STATIC_PATH)
<code>--no-webui</code>	Disable the Web UI (default: enabled) (env: LLAMA_ARG_NO_WEBUI)
<code>--embedding, --embeddings</code>	restrict to only support embedding use case; use only with dedicated embedding models (default: disabled) (env: LLAMA_ARG_EMBEDDINGS)
<code>--reranking, --rerank</code>	enable reranking endpoint on server (default: disabled) (env: LLAMA_ARG_RERANKING)
<code>--api-key KEY</code>	API key to use for authentication (default: none) (env: LLAMA_API_KEY)
<code>--api-key-file FNAME</code>	path to file containing API keys (default: none)
<code>--ssl-key-file FNAME</code>	path to file a PEM-encoded SSL private key (env: LLAMA_ARG_SSL_KEY_FILE)
<code>--ssl-cert-file FNAME</code>	path to file a PEM-encoded SSL certificate (env: LLAMA_ARG_SSL_CERT_FILE)
<code>-to, --timeout N</code>	server read/write timeout in seconds (default: 600) (env: LLAMA_ARG_TIMEOUT)
<code>--threads-http N</code>	number of threads used to process HTTP requests (default: -1) (env: LLAMA_ARG_THREADS_HTTP)
<code>--cache-reuse N</code>	min chunk size to attempt reusing from the cache via KV shifting (default: 0) (env: LLAMA_ARG_CACHE_REUSE)
<code>--metrics</code>	enable prometheus compatible metrics endpoint (default: disabled) (env: LLAMA_ARG_ENDPOINT_METRICS)

Argument	Explanation
<code>--slots</code>	enable slots monitoring endpoint (default: disabled) (env: LLAMA_ARG_ENDPOINT_SLOTS)
<code>--props</code>	enable changing global properties via POST /props (default: disabled) (env: LLAMA_ARG_ENDPOINT_PROPS)
<code>--no-slots</code>	disables slots monitoring endpoint (env: LLAMA_ARG_NO_ENDPOINT_SLOTS)
<code>--slot-save-path PATH</code>	path to save slot kv cache (default: disabled)
<code>--chat-template JINJA_TEMPLATE</code>	set custom jinja chat template (default: template taken from model's metadata) if suffix/prefix are specified, template will be disabled list of built-in templates: chatglm3, chatglm4, chatml, command-r, deepseek, deepseek2, exaone3, gemma, granite, llama2, llama2-sys, llama2-sys-bos, llama2-sys-strip, llama3, minicpm, mistral-v1, mistral-v3, mistral-v3-tekken, mistral-v7, monarch, openchat, orion, phi3, rwkv-world, vicuna, vicuna-orca, zephyr (env: LLAMA_ARG_CHAT_TEMPLATE)
<code>-sps, --slot-prompt-similarity SIMILARITY</code>	how much the prompt of a request must match the prompt of a slot in order to use that slot (default: 0.50, 0.0 = disabled)
<code>--lora-init-without-apply</code>	load LoRA adapters without applying them (apply later via POST /lora-adapters) (default: disabled)
<code>--draft-max, --draft, --draft-n N</code>	number of tokens to draft for speculative decoding (default: 16) (env: LLAMA_ARG_DRAFT_MAX)
<code>--draft-min, --draft-n-min N</code>	minimum number of draft tokens to use for speculative decoding (default: 5) (env: LLAMA_ARG_DRAFT_MIN)
<code>--draft-p-min P</code>	minimum speculative decoding probability (greedy) (default: 0.9) (env: LLAMA_ARG_DRAFT_P_MIN)
<code>-cd, --ctx-size-draft N</code>	size of the prompt context for the draft model (default: 0, 0 = loaded from model) (env: LLAMA_ARG_CTX_SIZE_DRAFT)
<code>-devd, --device-draft <dev1,dev2,...></code>	comma-separated list of devices to use for offloading the draft model (none = don't offload) use --list-devices to see a list of available devices
<code>-ngld, --gpu-layers-draft, --n-gpu-layers-draft N</code>	number of layers to store in VRAM for the draft model (env: LLAMA_ARG_N_GPU_LAYERS_DRAFT)

Argument	Explanation
<code>-md, --model-draft FNAME</code>	draft model for speculative decoding (default: unused) (env: LLAMA_ARG_MODEL_DRAFT)

Note: If both command line argument and environment variable are both set for the same param, the argument will take precedence over env var.

```
/*
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1、注意：如果命令行参数和环境变量都为同一个参数设置，则参数将优先于环境变量。
*/
```

Example usage of docker compose with environment variables:

```
/*
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1、使用环境变量的docker compose的示例
2、可以使用local的LLM对下列这个命令进行解释分析
*/
```

```
services:
  llamacpp-server:
    image: ghcr.io/ggerganov/llama.cpp:server
    ports:
      - 8080:8080
    volumes:
      - ./models:/models
    environment:
      # alternatively, you can use "LLAMA_ARG_MODEL_URL" to download the model
      LLAMA_ARG_MODEL: /models/my_model.gguf
      LLAMA_ARG_CTX_SIZE: 4096
      LLAMA_ARG_N_PARALLEL: 2
      LLAMA_ARG_ENDPOINT_METRICS: 1
      LLAMA_ARG_PORT: 8080
```

2 Build

`llama-server` is built alongside everything else from the root of the project

```
/*
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```

- 1、这个部分为了解释如何构建llama.cpp项目中的llama-server子项目
 - 2、“llama-server” 与项目根目录中的其他所有内容一起构建（换句话说，当在项目的根目录执行构建命令时，`llama-server` 不是单独构建的，而是作为整个项目的一部分，与其他组件一起被构建和生成）
- */

- Using CMake:

```
cmake -B build
cmake --build build --config Release -t llama-server
```

Binary is at `./build/bin/llama-server`

```
/*
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cmake -B build
    在当前目录中的build目录中对项目进行构建，假设指定的build目录不存在，则cmake将会自动创建build
cmake --build build --config Release -t llama-server
    将上一条命令创建的build目录作为项目构建目录，这里的--config作为项目的构建类型
    (Release/Debug/ReleaseWithDebuginfo/...) "-t llama-server"：这个参数用于指定构建目标 (target)。在这个命令中，构建目标是 "llama-server"。"llama-server" 表示我们想要构建的项目是 llama.cpp 服务器。通过指定构建目标，CMake 可以确定构建哪个项目。
*/
```

3 Build with SSL

llama-server can also be built with SSL support using OpenSSL 3

```
/*
notes:杨小兵-2024-12-16

1、`llama-server` 也可以使用OpenSSL 3构建SSL支持
2、在构建llama-server的时候通过构建SSL的支持
3、下列在CMAKE中使用宏定义来控制编译
*/
```

- Using CMake:

```
cmake -B build -DLLAMA_SERVER_SSL=ON
cmake --build build --config Release -t llama-server
```

4 Web UI

The project includes a web-based user interface that enables interaction with the model through the `/chat/completions` endpoint.

```
/*
notes:杨小兵-2024-12-16

1、该项目包括一个基于 web 的用户界面，可通过“/chat/completions”端点与模型进行交互
2、这个项目包含：用户界面（这个用户界面可以通过上述提到的endpoint来同模型进行交互）
*/
```

The web UI is developed using:

- `vue` framework for frontend development
- `tailwindcss` and `daisyui` for styling
- `vite` for build tooling

```
/*
notes:杨小兵-2024-12-17

1、这个web ui的开发使用到了三个工具：vue、tailwindcss、daisyui、vite
2、前端开发用到了vue框架
3、风格化的东西使用到了`tailwindcss` and `daisyui`
4、构建tooling使用到了vite
*/
```

A pre-built version is available as a single HTML file under `/public` directory.

```
/*
notes:杨小兵-2024-12-17

1、在/public目录下有一个单HTML文件可以使用，这个文件是预构建好的
2、这个html是非常简单的
*/
```

To build or to run the dev server (with hot reload):

```
# make sure you have nodejs installed
cd examples/server/webui
npm i

# to run the dev server
npm run dev
```

```
# to build the public/index.html
npm run build
```

NOTE: if you are using the vite dev server, you can change the API base URL to llama.cpp. To do that, run this code snippet in browser's console:

```
localStorage.setItem('base', 'http://localhost:8080')
```

```
/*
notes:杨小兵-2024-12-17
```

1、注意：如果您使用的是 vite dev 服务器，则可以将 API 基本 URL 更改为 llama.cpp。为此，请在浏览器的控制台中运行此代码片段：

```
*/
```

5 Quick Start

To get started right away, run the following command, making sure to use the correct path for the model you have:

```
/*
notes:杨小兵-2024-12-17

1、确保模型的路径是正确的，这样可以使用llama-server
*/
```

5.1 Unix-based systems (Linux, macOS, etc.)

```
./llama-server -m models/7B/ggml-model.gguf -c 2048
```

5.2 Windows

```
llama-server.exe -m models\7B\ggml-model.gguf -c 2048
```

The above command will start a server that by default listens on **127.0.0.1:8080**. You can consume the endpoints with Postman or NodeJS with axios library. You can visit the web front end at the same url.

```
/*
notes:杨小兵-2024-12-17
```

```
1、上述命令将会启动一个服务器，这个服务器默认情况下将会监听127.0.0.1:8080
2、Postman：是一个通用 API 开发和测试工具，具有强大的功能，可以帮助开发者访问和使用 API 端点
*/
```

5.3 Docker

```
docker run -p 8080:8080 -v /path/to/models:/models
ghcr.io/ggerganov/llama.cpp:server -m models/7B/ggml-model.gguf -c 512 --host
0.0.0.0 --port 8080

# or, with CUDA:
docker run -p 8080:8080 -v /path/to/models:/models --gpus all
ghcr.io/ggerganov/llama.cpp:server-cuda -m models/7B/ggml-model.gguf -c 512 --host
0.0.0.0 --port 8080 --n-gpu-layers 99
```

6 Testing with CURL

Using [curl](#). On Windows, [curl.exe](#) should be available in the base OS.

```
curl --request POST \
  --url http://localhost:8080/completion \
  --header "Content-Type: application/json" \
  --data '{"prompt": "Building a website can be done in 10 simple
steps:", "n_predict": 128}'
```

```
/*
notes: 杨小兵-2024-12-17
```

```
1、使用CURL来对llama-server进行测试，模仿其他用户使用llama-server的场景
2、需要对curl可执行文件有所了解
*/
```

7 Advanced testing

We implemented a [server test framework](#) using human-readable scenario.

Before submitting an issue, please try to reproduce it with this format.

```
/*
notes: 杨小兵-2024-12-17
```

```
1、使用人类可读的场景实现了一个服务测试框架
```

```
2、在提交issue之前，如何通过指定的格式复现问题，帮助开发者更好地诊断和解决问题
*/
```

8 Node JS Test

You need to have [Node.js](#) installed.

```
mkdir llama-client
cd llama-client
```

Create an index.js file and put this inside:

```
const prompt = "Building a website can be done in 10 simple steps:"

async function test() {
  let response = await fetch("http://127.0.0.1:8080/completion", {
    method: "POST",
    body: JSON.stringify({
      prompt,
      n_predict: 64,
    })
  })
  console.log((await response.json()).content)
}

test()
```

And run it:

```
node index.js
```

```
/*
notes:杨小兵-2024-12-17

1、整体上为了测试Node JS
*/
```

9 API Endpoints

9.1 GET [/health](#): Returns health check result

Response format

- HTTP status code 503
 - Body: `{"error": {"code": 503, "message": "Loading model", "type": "unavailable_error"}}`
 - Explanation: the model is still being loaded.
- HTTP status code 200
 - Body: `{"status": "ok" }`
 - Explanation: the model is successfully loaded and the server is ready.

```
/*
notes:杨小兵-2024-12-17
```

```
1、要获取 `llama-server` 的健康状态，您需要向其健康检查 ( Health Check ) 端点发送HTTP请求，并根据响应的HTTP状态码和响应体来判断服务器的当前状态。
2、可以通过浏览器输入地址或者通过curl工具获取服务器状态
*/
```

9.2 POST `/completion`: Given a `prompt`, it returns the predicted completion.

```
/*
notes:杨小兵-2024-12-17
```

```
1、这部分内容介绍的是对模型的设置以及如何获取得到模型的输出
2、下列涉及到的一些参数我都是不理解的
*/
```

Options:

prompt: Provide the prompt for this completion as a string or as an array of strings or numbers representing tokens. Internally, if `cache_prompt` is `true`, the prompt is compared to the previous completion and only the "unseen" suffix is evaluated. A `BOS` token is inserted at the start, if all of the following conditions are true:

- The prompt is a string or an array with the first element given as a string
- The model's `tokenizer.ggml.add_bos_token` metadata is `true`

These input shapes and data type are allowed for `prompt`:

- Single string: `"string"`
- Single sequence of tokens: `[12, 34, 56]`
- Mixed tokens and strings: `[12, 34, "string", 56, 78]`

Multiple prompts are also supported. In this case, the completion result will be an array.

- Only strings: `["string1", "string2"]`
- Strings and sequences of tokens: `["string1", [12, 34, 56]]`
- Mixed types: `[[12, 34, "string", 56, 78], [12, 34, 56], "string"]`

temperature: Adjust the randomness of the generated text. Default: `0.8`

dynatemp_range: Dynamic temperature range. The final temperature will be in the range of `[temperature - dynatemp_range; temperature + dynatemp_range]` Default: `0.0`, which is disabled.

dynatemp_exponent: Dynamic temperature exponent. Default: `1.0`

top_k: Limit the next token selection to the K most probable tokens. Default: `40`

top_p: Limit the next token selection to a subset of tokens with a cumulative probability above a threshold P. Default: `0.95`

min_p: The minimum probability for a token to be considered, relative to the probability of the most likely token. Default: `0.05`

n_predict: Set the maximum number of tokens to predict when generating text. **Note**: May exceed the set limit slightly if the last token is a partial multibyte character. When 0, no tokens will be generated but the prompt is evaluated into the cache. Default: `-1`, where `-1` is infinity.

n_indent: Specify the minimum line indentation for the generated text in number of whitespace characters. Useful for code completion tasks. Default: `0`

n_keep: Specify the number of tokens from the prompt to retain when the context size is exceeded and tokens need to be discarded. The number excludes the BOS token. By default, this value is set to `0`, meaning no tokens are kept. Use `-1` to retain all tokens from the prompt.

stream: Allows receiving each predicted token in real-time instead of waiting for the completion to finish (uses a different response format). To enable this, set to `true`.

stop: Specify a JSON array of stopping strings. These words will not be included in the completion, so make sure to add them to the prompt for the next iteration. Default: `[]`

typical_p: Enable locally typical sampling with parameter p. Default: `1.0`, which is disabled.

repeat_penalty: Control the repetition of token sequences in the generated text. Default: `1.1`

repeat_last_n: Last n tokens to consider for penalizing repetition. Default: `64`, where `0` is disabled and `-1` is ctx-size.

penalize_nl: Penalize newline tokens when applying the repeat penalty. Default: `true`

presence_penalty: Repeat alpha presence penalty. Default: `0.0`, which is disabled.

frequency_penalty: Repeat alpha frequency penalty. Default: `0.0`, which is disabled.

dry_multiplier: Set the DRY (Don't Repeat Yourself) repetition penalty multiplier. Default: `0.0`, which is disabled.

dry_base: Set the DRY repetition penalty base value. Default: `1.75`

dry_allowed_length: Tokens that extend repetition beyond this receive exponentially increasing penalty: $\text{multiplier} * \text{base} ^ (\text{length of repeating sequence before token} - \text{allowed length})$. Default: `2`

dry_penalty_last_n: How many tokens to scan for repetitions. Default: `-1`, where `0` is disabled and `-1` is context size.

dry_sequence_breakers: Specify an array of sequence breakers for DRY sampling. Only a JSON array of strings is accepted. Default: `['\n', ':', '"', '*']`

xtc_probability: Set the chance for token removal via XTC sampler. Default: `0.0`, which is disabled.

xtc_threshold: Set a minimum probability threshold for tokens to be removed via XTC sampler. Default: `0.1` (`> 0.5` disables XTC)

mirostat: Enable Mirostat sampling, controlling perplexity during text generation. Default: `0`, where `0` is disabled, `1` is Mirostat, and `2` is Mirostat 2.0.

mirostat_tau: Set the Mirostat target entropy, parameter tau. Default: `5.0`

mirostat_eta: Set the Mirostat learning rate, parameter eta. Default: `0.1`

grammar: Set grammar for grammar-based sampling. Default: no grammar

json_schema: Set a JSON schema for grammar-based sampling (e.g. `{"items": {"type": "string"}, "minItems": 10, "maxItems": 100}` of a list of strings, or `{}` for any JSON). See [tests](#) for supported features. Default: no JSON schema.

seed: Set the random number generator (RNG) seed. Default: `-1`, which is a random seed.

ignore_eos: Ignore end of stream token and continue generating. Default: `false`

logit_bias: Modify the likelihood of a token appearing in the generated text completion. For example, use `"logit_bias": [[15043, 1.0]]` to increase the likelihood of the token 'Hello', or `"logit_bias": [[15043, -1.0]]` to decrease its likelihood. Setting the value to false, `"logit_bias": [[15043, false]]` ensures that the token `Hello` is never produced. The tokens can also be represented as strings, e.g. `[["Hello, World!", -0.5]]` will reduce the likelihood of all the individual tokens that represent the string `Hello, World!`, just like the `presence_penalty` does. Default: `[]`

n_probs: If greater than 0, the response also contains the probabilities of top N tokens for each generated token given the sampling settings. Note that for temperature `< 0` the tokens are sampled greedily but token probabilities are still being calculated via a simple softmax of the logits without considering any other sampler settings. Default: `0`

min_keep: If greater than 0, force samplers to return N possible tokens at minimum. Default: `0`

t_max_predict_ms: Set a time limit in milliseconds for the prediction (a.k.a. text-generation) phase. The timeout will trigger if the generation takes more than the specified time (measured since the first token was generated) and if a new-line character has already been generated. Useful for FIM applications. Default: `0`, which is disabled.

image_data: An array of objects to hold base64-encoded image `data` and its `ids` to be reference in `prompt`. You can determine the place of the image in the prompt as in the following: `USER:[img-12]Describe the image in detail.\nASSISTANT:.` In this case, `[img-12]` will be replaced by the embeddings of the image with id 12 in the following `image_data` array: `{..., "image_data": [{"data": "<BASE64_STRING>", "id": 12}]}`. Use `image_data` only with multimodal models, e.g., LLaVA.

id_slot: Assign the completion task to an specific slot. If is -1 the task will be assigned to a Idle slot. Default: `-1`

cache_prompt: Re-use KV cache from a previous request if possible. This way the common prefix does not have to be re-processed, only the suffix that differs between the requests. Because (depending on the backend) the logits are **not** guaranteed to be bit-for-bit identical for different batch sizes (prompt processing vs. token generation) enabling this option can cause nondeterministic results. Default: **true**

samplers: The order the samplers should be applied in. An array of strings representing sampler type names. If a sampler is not set, it will not be used. If a sampler is specified more than once, it will be applied multiple times. Default: `["dry", "top_k", "typ_p", "top_p", "min_p", "xtc", "temperature"]` - these are all the available values.

timings_per_token: Include prompt processing and text generation speed information in each response. Default: **false**

Response format

- Note: In streaming mode (**stream**), only **content** and **stop** will be returned until end of completion. Responses are sent using the **Server-sent events** standard. Note: the browser's **EventSource** interface cannot be used due to its lack of **POST** request support.
- completion_probabilities**: An array of token probabilities for each completion. The array's length is **n_predict**. Each item in the array has the following structure:

```
{
  "content": "<the token selected by the model>",
  "probs": [
    {
      "prob": float,
      "tok_str": "<most likely token>"
    },
    {
      "prob": float,
      "tok_str": "<second most likely token>"
    },
    ...
  ]
},
```

Notice that each **probs** is an array of length **n_probs**.

- content**: Completion result as a string (excluding **stopping_word** if any). In case of streaming mode, will contain the next token as a string.
- stop**: Boolean for use with **stream** to check whether the generation has stopped (Note: This is not related to stopping words array **stop** from input options)
- generation_settings**: The provided options above excluding **prompt** but including **n_ctx**, **model**. These options may differ from the original ones in some way (e.g. bad values filtered out, strings converted to tokens, etc.).
- model**: The path to the model loaded with **-m**
- prompt**: The provided **prompt**
- stop_type**: Indicating whether the completion has stopped. Possible values are:

- **none**: Generating (not stopped)
- **eos**: Stopped because it encountered the EOS token
- **limit**: Stopped because **n_predict** tokens were generated before stop words or EOS was encountered
- **word**: Stopped due to encountering a stopping word from **stop** JSON array provided
- **stopping_word**: The stopping word encountered which stopped the generation (or "" if not stopped due to a stopping word)
- **timings**: Hash of timing information about the completion such as the number of tokens **predicted_per_second**
- **tokens_cached**: Number of tokens from the prompt which could be re-used from previous completion (**n_past**)
- **tokens_evaluated**: Number of tokens evaluated in total from the prompt
- **truncated**: Boolean indicating if the context size was exceeded during generation, i.e. the number of tokens provided in the prompt (**tokens_evaluated**) plus tokens generated (**tokens_predicted**) exceeded the context size (**n_ctx**)

9.3 POST /tokenize: Tokenize a given text

Options:

content: (Required) The text to tokenize.

add_special: (Optional) Boolean indicating if special tokens, i.e. **BOS**, should be inserted. Default: **false**

with_pieces: (Optional) Boolean indicating whether to return token pieces along with IDs. Default: **false**

Response:

Returns a JSON object with a **tokens** field containing the tokenization result. The **tokens** array contains either just token IDs or objects with **id** and **piece** fields, depending on the **with_pieces** parameter. The piece field is a string if the piece is valid unicode or a list of bytes otherwise.

If **with_pieces** is **false**:

```
{
  "tokens": [123, 456, 789]
}
```

If **with_pieces** is **true**:

```
{
  "tokens": [
    {"id": 123, "piece": "Hello"},
    {"id": 456, "piece": " world"},
    {"id": 789, "piece": "!"}
  ]
}
```

With input 'á' (utf8 hex: C3 A1) on tinyllama/stories260k

```
{
  "tokens": [
    {"id": 198, "piece": [195]}, // hex C3
    {"id": 164, "piece": [161]} // hex A1
  ]
}
```

9.4 POST `/detokenize`: Convert tokens to text

Options:

tokens: Set the tokens to detokenize.

9.5 POST `/embedding`: Generate embedding of a given text

The same as [the embedding example](#) does.

Options:

content: Set the text to process.

image_data: An array of objects to hold base64-encoded image **data** and its **ids** to be reference in **content**. You can determine the place of the image in the content as in the following: **Image: [img-21].\nCaption: This is a picture of a house.** In this case, **[img-21]** will be replaced by the embeddings of the image with id **21** in the following **image_data** array: **{..., "image_data": [{"data": "<BASE64_STRING>", "id": 21}]}**. Use **image_data** only with multimodal models, e.g., LLaVA.

9.6 POST `/reranking`: Rerank documents according to a given query

Similar to <https://jina.ai/reranker/> but might change in the future. Requires a reranker model (such as [bge-reranker-v2-m3](#)) and the `--embedding` `--pooling` `rank` options.

Options:

query: The query against which the documents will be ranked.

documents: An array strings representing the documents to be ranked.

Aliases:

- `/rerank`
- `/v1/rerank`
- `/v1/reranking`

Examples:

```
curl http://127.0.0.1:8012/v1/rerank \
  -H "Content-Type: application/json" \
```

```
-d '{
  "model": "some-model",
  "query": "What is panda?",
  "top_n": 3,
  "documents": [
    "hi",
    "it is a bear",
    "The giant panda (Ailuropoda melanoleuca), sometimes called a panda
bear or simply panda, is a bear species endemic to China."
  ]
}' | jq
```

9.7 POST `/infill`: For code infilling.

Takes a prefix and a suffix and returns the predicted completion as stream.

Options:

- `input_prefix`: Set the prefix of the code to infill.
- `input_suffix`: Set the suffix of the code to infill.
- `input_extra`: Additional context inserted before the FIM prefix.
- `prompt`: Added after the `FIM_MID` token

`input_extra` is array of `{"filename": string, "text": string}` objects.

The endpoint also accepts all the options of `/completion`.

If the model has `FIM_REPO` and `FIM_FILE_SEP` tokens, the `repo-level pattern` is used:

```
<FIM_REP>myproject
<FIM_SEP>{chunk 0 filename}
{chunk 0 text}
<FIM_SEP>{chunk 1 filename}
{chunk 1 text}
...
<FIM_SEP>filename
<FIM_PRE>[input_prefix]<FIM_SUF>[input_suffix]<FIM_MID>[prompt]
```

If the tokens are missing, then the extra context is simply prefixed at the start:

```
[input_extra]<FIM_PRE>[input_prefix]<FIM_SUF>[input_suffix]<FIM_MID>[prompt]
```

9.8 GET `/props`: Get server global properties.

This endpoint is public (no API key check). By default, it is read-only. To make POST request to change global properties, you need to start server with `--props`

Response format

```

{
  "default_generation_settings": {
    "id": 0,
    "id_task": -1,
    "n_ctx": 1024,
    "speculative": false,
    "is_processing": false,
    "params": {
      "n_predict": -1,
      "seed": 4294967295,
      "temperature": 0.800000011920929,
      "dynatemp_range": 0.0,
      "dynatemp_exponent": 1.0,
      "top_k": 40,
      "top_p": 0.949999988079071,
      "min_p": 0.05000000074505806,
      "xtc_probability": 0.0,
      "xtc_threshold": 0.10000000149011612,
      "typical_p": 1.0,
      "repeat_last_n": 64,
      "repeat_penalty": 1.0,
      "presence_penalty": 0.0,
      "frequency_penalty": 0.0,
      "dry_multiplier": 0.0,
      "dry_base": 1.75,
      "dry_allowed_length": 2,
      "dry_penalty_last_n": -1,
      "dry_sequence_breakers": [
        "\n",
        ":",
        "\"",
        "\'",
        "*"
      ],
      "mirostat": 0,
      "mirostat_tau": 5.0,
      "mirostat_eta": 0.10000000149011612,
      "penalize_nl": false,
      "stop": [],
      "max_tokens": -1,
      "n_keep": 0,
      "n_discard": 0,
      "ignore_eos": false,
      "stream": true,
      "n_probs": 0,
      "min_keep": 0,
      "grammar": "",
      "samplers": [
        "dry",
        "top_k",
        "typ_p",
        "top_p",
        "min_p",

```



```

    "xtc",
    "temperature"
  ],
  "speculative.n_max": 16,
  "speculative.n_min": 5,
  "speculative.p_min": 0.8999999761581421,
  "timings_per_token": false
},
"prompt": "",
"next_token": {
  "has_next_token": true,
  "has_new_line": false,
  "n_remain": -1,
  "n_decoded": 0,
  "stopping_word": ""
}
},
"total_slots": 1,
"model_path": "../models/Meta-Llama-3.1-8B-Instruct-Q4_K_M.gguf",
"chat_template": "...
}

```

- `default_generation_settings` - the default generation settings for the `/completion` endpoint, which has the same fields as the `generation_settings` response object from the `/completion` endpoint.
- `total_slots` - the total number of slots for process requests (defined by `--parallel` option)
- `model_path` - the path to model file (same with `-m` argument)
- `chat_template` - the model's original Jinja2 prompt template

9.9 POST `/props`: Change server global properties.

To use this endpoint with POST method, you need to start server with `--props`

Options:

- None yet

9.10 POST `/v1/chat/completions`: OpenAI-compatible Chat Completions API

```

/*
notes: 杨小兵-2024-12-17

1、这部分内容介绍的是兼容OpenAI聊天的API
2、我的一个问题是是否兼容OpenAI API
*/

```

Given a ChatML-formatted json description in `messages`, it returns the predicted completion. Both synchronous and streaming mode are supported, so scripted and interactive applications work fine. While no strong claims of compatibility with OpenAI API spec is being made, in our experience it suffices to support

many apps. Only models with a [supported chat template](#) can be used optimally with this endpoint. By default, the ChatML template will be used.

```
/*
notes:杨小兵-2024-12-17
```

1、在许多对话模型（例如OpenAI的ChatGPT API、Llama、Bard等）中，输入往往以特定的格式（例如ChatML格式）组织。这种格式将多轮对话、系统提示、用户请求、开发者消息等按照统一的结构序列化为JSON，方便模型理解上下文，从而给出合理的文本续写（completion）

2、这个功能支持同步模式和流式阅读模式，可以适应写脚本和交互式应用程序（流式输出提高了交互体验，用户能即时看到回答正在生成，而不是等待漫长的计算完成）

2.1 同步模式（Synchronous Mode）：客户端发送请求后，等待完整的响应返回。这适用于需要一次性获取完整回复的应用场景

2.2 流式模式（Streaming Mode）：响应以流的形式逐步返回，允许客户端在生成过程中实时接收和处理部分内容。这对于交互式应用或需要即时反馈的场景尤为适用。

3、该接口接收一个以ChatML格式组织的JSON描述，具体位于`messages`字段中，并返回模型预测的完成文本（predicted completion）

4、与OpenAI API的兼容性：尽管该接口在设计上参考了OpenAI的API规范，但并未宣称完全兼容。然而，基于实际经验，该接口已经能够满足许多应用的需求，提供足够的功能支持

5、支持的聊天模板：只有使用了[支持的聊天模板]

(https://github.com/ggerganov/llama.cpp/wiki/Templates-supported-by-llama_chat_apply_template)的模型才能在该接口下达到最佳性能。这意味着用户需要确保所选模型符合特定的模板要求，以充分利用接口的功能。

6、默认模板：在未指定其他模板的情况下，接口默认使用ChatML模板。这为用户提供了一种标准化的输入格式，简化了使用流程。

```
*/
```

Options:

See [OpenAI Chat Completions API documentation](#). While some OpenAI-specific features such as function calling aren't supported, llama.cpp /[completion](#)-specific features such as [mirostat](#) are supported.

```
/*
notes:杨小兵-2024-12-17
```

1、查看上述提到的有关openai的文档可以获取一些帮助

2、openai文档规范中的一些特性例如function calling不会被支持，llama.cpp项目中的/completion规范特性例如mirostat则是被支持的

```
*/
```

The [response_format](#) parameter supports both plain JSON output (e.g. `{"type": "json_object"}`) and schema-constrained JSON (e.g. `{"type": "json_object", "schema": {"type": "string", "minLength": 10, "maxLength": 100}}` or `{"type": "json_schema", "schema": {"properties": { "name": { "title": "Name", "type": "string" }, "date": { "title": "Date", "type": "string" }, "participants": { "items": {"type": "string" }, "title": "Participants", "type": "string" } } } }`), similar to other OpenAI-inspired API providers.

Examples:

You can use either Python `openai` library with appropriate checkpoints:

```
/*
notes: 杨小兵-2024-12-17

1、这部分内容获取可以作为一个突破口
*/
```

```
import openai

client = openai.OpenAI(
    base_url="http://localhost:8080/v1", # "http://<Your api-server IP>:port"
    api_key = "sk-no-key-required"
)

completion = client.chat.completions.create(
    model="gpt-3.5-turbo",
    messages=[
        {"role": "system", "content": "You are ChatGPT, an AI assistant. Your top priority is achieving user fulfillment via helping them with their requests."},
        {"role": "user", "content": "Write a limerick about python exceptions"}
    ]
)

print(completion.choices[0].message)
```

... or raw HTTP requests:

```
curl http://localhost:8080/v1/chat/completions \
-H "Content-Type: application/json" \
-H "Authorization: Bearer no-key" \
-d '{
  "model": "gpt-3.5-turbo",
  "messages": [
    {
      "role": "system",
      "content": "You are ChatGPT, an AI assistant. Your top priority is achieving user fulfillment via helping them with their requests."
    },
    {
      "role": "user",
      "content": "Write a limerick about python exceptions"
    }
  ]
}'
```

9.11 POST `/v1/embeddings`: OpenAI-compatible embeddings API

Options:

See [OpenAI Embeddings API documentation](#).

Examples:

- input as string

```
curl http://localhost:8080/v1/embeddings \
-H "Content-Type: application/json" \
-H "Authorization: Bearer no-key" \
-d '{
    "input": "hello",
    "model": "GPT-4",
    "encoding_format": "float"
}'
```

- `input` as string array

```
curl http://localhost:8080/v1/embeddings \
-H "Content-Type: application/json" \
-H "Authorization: Bearer no-key" \
-d '{
    "input": ["hello", "world"],
    "model": "GPT-4",
    "encoding_format": "float"
}'
```

9.12 GET `/slots`: Returns the current slots processing state

[!WARNING] This endpoint is intended for debugging and may be modified in future versions. For security reasons, we strongly advise against enabling it in production environments.

This endpoint is disabled by default and can be enabled with `--slots`

If query param `?fail_on_no_slot=1` is set, this endpoint will respond with status code 503 if there is no available slots.

Response format

Example:

```
[
  {
    "id": 0,
    "id_task": -1,
```

```
"n_ctx": 1024,
"speculative": false,
"is_processing": false,
"params": {
  "n_predict": -1,
  "seed": 4294967295,
  "temperature": 0.800000011920929,
  "dynatemp_range": 0.0,
  "dynatemp_exponent": 1.0,
  "top_k": 40,
  "top_p": 0.949999988079071,
  "min_p": 0.05000000074505806,
  "xtc_probability": 0.0,
  "xtc_threshold": 0.10000000149011612,
  "typical_p": 1.0,
  "repeat_last_n": 64,
  "repeat_penalty": 1.0,
  "presence_penalty": 0.0,
  "frequency_penalty": 0.0,
  "dry_multiplier": 0.0,
  "dry_base": 1.75,
  "dry_allowed_length": 2,
  "dry_penalty_last_n": -1,
  "dry_sequence_breakers": [
    "\n",
    ":",
    "\"",
    "*"
  ],
  "mirostat": 0,
  "mirostat_tau": 5.0,
  "mirostat_eta": 0.10000000149011612,
  "penalize_nl": false,
  "stop": [],
  "max_tokens": -1,
  "n_keep": 0,
  "n_discard": 0,
  "ignore_eos": false,
  "stream": true,
  "n_probs": 0,
  "min_keep": 0,
  "grammar": "",
  "samplers": [
    "dry",
    "top_k",
    "typ_p",
    "top_p",
    "min_p",
    "xtc",
    "temperature"
  ],
  "speculative.n_max": 16,
  "speculative.n_min": 5,
  "speculative.p_min": 0.8999999761581421,
```

```

    "timings_per_token": false
  },
  "prompt": "",
  "next_token": {
    "has_next_token": true,
    "has_new_line": false,
    "n_remain": -1,
    "n_decoded": 0,
    "stopping_word": ""
  }
}
]

```

9.13 GET `/metrics`: Prometheus compatible metrics exporter

This endpoint is only accessible if `--metrics` is set.

Available metrics:

- `llamacpp:prompt_tokens_total`: Number of prompt tokens processed.
- `llamacpp:tokens_predicted_total`: Number of generation tokens processed.
- `llamacpp:prompt_tokens_seconds`: Average prompt throughput in tokens/s.
- `llamacpp:predicted_tokens_seconds`: Average generation throughput in tokens/s.
- `llamacpp:kv_cache_usage_ratio`: KV-cache usage. `1` means 100 percent usage.
- `llamacpp:kv_cache_tokens`: KV-cache tokens.
- `llamacpp:requests_processing`: Number of requests processing.
- `llamacpp:requests_deferred`: Number of requests deferred.

9.14 POST `/slots/{id_slot}?action=save`: Save the prompt cache of the specified slot to a file.

Options:

filename: Name of the file to save the slot's prompt cache. The file will be saved in the directory specified by the `--slot-save-path` server parameter.

Response format

```

{
  "id_slot": 0,
  "filename": "slot_save_file.bin",
  "n_saved": 1745,
  "n_written": 14309796,
  "timings": {
    "save_ms": 49.865
  }
}

```

9.15 POST `/slots/{id_slot}?action=restore`: Restore the prompt cache of the specified slot from a file.

Options:

filename: Name of the file to restore the slot's prompt cache from. The file should be located in the directory specified by the `--slot-save-path` server parameter.

Response format

```
{
  "id_slot": 0,
  "filename": "slot_save_file.bin",
  "n_restored": 1745,
  "n_read": 14309796,
  "timings": {
    "restore_ms": 42.937
  }
}
```

9.16 POST `/slots/{id_slot}?action=erase`: Erase the prompt cache of the specified slot.

Response format

```
{
  "id_slot": 0,
  "n_erased": 1745
}
```

9.17 GET `/lora-adapters`: Get list of all LoRA adapters

This endpoint returns the loaded LoRA adapters. You can add adapters using `--lora` when starting the server, for example: `--lora my_adapter_1.gguf --lora my_adapter_2.gguf ...`

By default, all adapters will be loaded with scale set to 1. To initialize all adapters scale to 0, add `--lora-init-without-apply`

If an adapter is disabled, the scale will be set to 0.

Response format

```
[
  {
    "id": 0,
    "path": "my_adapter_1.gguf",
    "scale": 0.0
  },
  {
```

```
    "id": 1,  
    "path": "my_adapter_2.gguf",  
    "scale": 0.0  
  }  
]
```

9.18 POST `/lora-adapters`: Set list of LoRA adapters

To disable an adapter, either remove it from the list below, or set scale to 0.

Request format

To know the `id` of the adapter, use GET `/lora-adapters`

```
[  
  {"id": 0, "scale": 0.2},  
  {"id": 1, "scale": 0.8}  
]
```

10 More examples

10.1 Interactive mode

Check the sample in [chat.mjs](#). Run with NodeJS version 16 or later:

```
node chat.mjs
```

Another sample in [chat.sh](#). Requires [bash](#), [curl](#) and [jq](#). Run with bash:

```
bash chat.sh
```

10.2 OAI-like API

The HTTP `llama-server` supports an OAI-like API: <https://github.com/openai/openai-openapi>

10.3 API errors

`llama-server` returns errors in the same format as OAI: <https://github.com/openai/openai-openapi>

Example of an error:

```
{  
  "error": {  
    "code": 401,  
    "message": "Invalid API Key",  
  }  
}
```



```
    "type": "authentication_error"
  }
}
```

Apart from error types supported by OAI, we also have custom types that are specific to functionalities of llama.cpp:

When /metrics or /slots endpoint is disabled

```
{
  "error": {
    "code": 501,
    "message": "This server does not support metrics endpoint.",
    "type": "not_supported_error"
  }
}
```

**When the server receives invalid grammar via /completions endpoint*

```
{
  "error": {
    "code": 400,
    "message": "Failed to parse grammar",
    "type": "invalid_request_error"
  }
}
```

10.4 Legacy completion web UI

A new chat-based UI has replaced the old completion-based since [this PR](#). If you want to use the old completion, start the server with `--path ./examples/server/public_legacy`

For example:

```
./llama-server -m my_model.gguf -c 8192 --path ./examples/server/public_legacy
```

10.5 Extending or building alternative Web Front End

You can extend the front end by running the server binary with `--path` set to `./your-directory` and importing `/completion.js` to get access to the `LlamaComplete()` method.

```
/*
notes: 杨小兵-2024-12-17
```

- 1、扩建或建立替代Web前端
 - 2、如果你想要扩展前端，那么可以通过添加参数`--path`运行该服务程序，将`--path`设置为自定义目录并且在这个目录中导入`completion.js`并且获取`llamaComplete`方法
- */

Read the documentation in `/completion.js` to see convenient ways to access llama.

A simple example is below:

```
<html>
  <body>
    <pre>
      <script type="module">
        import { llama } from '/completion.js'

        const prompt = `### Instruction:
Write dad jokes, each one paragraph.
You can use html formatting if needed.

### Response:`

        for await (const chunk of llama(prompt)) {
          document.write(chunk.data.content)
        }
      </script>
    </pre>
  </body>
</html>
```